Homo Algebra by Rotman, Chapter 3

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Projective Modules Split Exact Sequences

P is projective $\implies$ every $0 \to A \to B \to P \to 0$ splits

$$\begin{array}{ccccccccc} & & & & & & P & & \\ & & & & & \swarrow & \downarrow{\scriptstyle 1_P} & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & P & \longrightarrow & 0 \end{array}$$

we induce $j : P \to B$ with $1_P = pj$ so that P is a retract of B.

Every $0 \to A \to B \to P \to 0$ splits $\implies P$ is projective

We can construct an exact sequence which splits by the hypothesis, then we have the projective map.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^m & \longrightarrow & R^n & \rightleftarrows & P & \longrightarrow & 0 \\ & & & & \downarrow & & \downarrow & & \\ & & & & B & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

Let A be a submodule of $B \in {}_R\text{Mod}$. If B/A is projective, then $B = A \oplus B/A$.

P is Projective $\iff$ P is a Direct Summand of a Free Module

$$0 \to \ker g \to F \to P \to 0$$

Projective Basis

$A \in {}_R\text{Mod}$ is projective $\iff \exists (a_i \in A)_{i\in I}$ and $(\phi_i : A \to R)_{i\in I}$ such that $\forall x \in A$, we have $x = \sum_{i\in I}(\phi_i x)a_i$.

Also $A = \langle (a_i) \rangle$.

Let A be projective, then $\psi : F \to A$ with ϕ by split property.

Define $a_i = \psi(e_i)$. For $x \in A$, then

$$\phi(x) = r_1 e_1 + \cdots + r_n e_n$$

Define $\phi_i(x) = r_i$, then

$$x = (\phi_1 x)a_1 + \cdots + (\phi_n x)a_n$$

For the converse, let there be $(a_i \in A)_{i\in I}$ and $(\phi_i : A \to R)_{i\in I}$. Define $\psi : F \to A$ by $\psi(e_i) = a_i$. Define $\phi(x) = (\phi_1 x)e_1 + \cdots + (\phi_n x)e_n$ then $\psi\phi = 1$. So A is a direct summand of F and therefore projective.

Schanuel's Lemma

$$0 \to K \to P \to M \to 0$$

$$0 \to K' \to P'M \to 0$$

with P, P' projective

$$\implies K \oplus P' \cong K' \oplus P$$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & P & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle 1_M} & & \\ 0 & \longrightarrow & K' & \xrightarrow{i'} & P' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

Since P is projective, we induce the middle map $P \to P'$. Then we get the left one for free.

Now we use this to construct

$$0 \longrightarrow K \xrightarrow{\theta} P \oplus K' \xrightarrow{\psi} P' \longrightarrow 0$$

which splits since P' is projective.

$$\theta(x) = (ix, \alpha x)$$

$$\psi(u, x') = \beta u - i'x'$$

Proposition 3.18/Corollary 3.19: Noetherian

Submodules remain finitely generated of left noetherian modules

Let R be left noetherian, and $M \in {}_R\text{Mod}$ be finitely generated. Then every submodule is finitely generated.

If $n = 1$, then $M = \langle m \rangle \cong R/I$ for some ideal. Let $S \subseteq M$ be the submodule, then $S \cong J/I$ for some ideal $I \subseteq J \subseteq R$. Since R is left noetherian, this means J and hence J/I is finitely generated, and so is S.

Now let $n \geq 1$ with $M = \langle x_1, ..., x_n, x_{n+1} \rangle$ and $M' = \langle x_1, ..., x_n \rangle$.

$$0 \to M' \to M \to M'' = M/M' \to 0$$

$M'' = \langle x_n + M' \rangle$ is cyclic. Let $S \subseteq M$ be the submodule, then

$$0 \to S \cap M' \to S \to S/(S \cap M') \to 0$$

Given any sequence $0 \to A \to B \to C \to 0$, if A and C are finitely generated, then so is B.

$S \cap M' \subseteq M'$ and so is finitely generated. $S/(S \cap M') \cong (S + M')/M' \subseteq M/M'$ and so is also finitely generated. Thus S is finitely generated.

Left noetherian module over PID ring generated by n or fewer elements

For the base step $n = 1$, M is therefore cyclic, and J is PID by hypothesis, so that J/I is cyclic too.

Same as above, but use that if A and C are generated by m and n elements, then B is generated by $m + n$ elements (or less).

$M \in {}_R$Mod where R is left noetherian is finitely presented

$\phi : F \to M$ is surjective, then by above all submodules including $\ker \phi$ are finitely generated.

Baer's Criterion

Use Zorn's lemma with partially ordered pairs fixing the previous item. The maximal element is (A_0, g_0). So there's a $b \in B : b \notin A_0$.

Let $I = \{r : rb \in A_0\}$. This is an ideal since A_0 is a submodule. It is all multiples of b that are in A_0. But there are some primitive multiples (and their factors) that remain outside A_0. Define the map $h : I \to E$

$$h(r) = g_0(rb)$$

By the hypothesis, this extends to $h^* : R \to E$. Then we can extend $A_1 = A_0 + \langle b \rangle$ and make $g_1 : A \to E$ by $g_1(a_0 + rb) = g_0(a_0) + rh^*(1)$. We can see g_1 extends g_0 which contradicts the maximality of (A_0, g_0).

Now for the final step, we must just prove that g_1 is well defined. Let $a_0 + rb = a_0' + r'b$.

- $(r - r')b = a_0' - a_0 \in A_0 \implies r - r' \in I$.
- Thus $g_0((r - r')b)$ and $h(r - r')$ are defined.
- $g_0((r - r')b) = (r - r')h^*(1) \implies g_0(r) + rh^*(1) = g_0(r') + r'h^*(1)$

So even if there are multiple ways of writing an element as $a_0 + rb$ then applying g_1 always gives the same result.

- Direct product of injective modules is injective.
- If R is left noetherian, then direct sum of injective R-modules is injective.
- If R is a domain then every injective R-module is divisible.
- If R is a domain, then Frac(Q) is an injective R-module.
- If R is PID, then E is injective $\iff E$ is divisible.
- If R is PID, then every quotient of injective modules are injective.
- If D is a divisible abeligan group, then $\text{Hom}_{\mathbb{Z}}(R, D)$ is an injective left R-module.

Every Module Can be Embedded in an Injective Module

This means injective sequences split since we can construct $0 \to M'' \to M \to E \to 0$ where M is injective.

$E \in {}_R$Mod is Injective $\iff$ every $0 \to E \to B \to C \to 0$ where C is Cyclic Splits

E is injective $\implies$ every sequence splits, so we only need to prove the converse.

Let I be an ideal of R, and let $f : I \to E$. By category (see prev lemma 3.41), we can construct this diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & R & \longrightarrow & R/I & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle 1_{R/I}} & & \\ 0 & \longrightarrow & E & \longrightarrow & P & \longrightarrow & R/I & \longrightarrow & 0 \end{array}$$

Since R/I is cyclic, by hypothesis the bottom row splits.

Thus there's a map $P \to E$ which we compose with $R \to P$ to get $R \to E$, then by Baer's criterion, we have shown E is injective.

Flat Modules

- R is a flat R-module.
- Direct sums are flat, and flat direct sums means all their submodules are flat.
- Projective modules are flat.
- Let $0 \to A \to B$ and $u \in \ker(1 \otimes i)$. Then there is a finitely generated $N \subseteq M$ and $u' \in N \otimes A$ such that:
  - $u' \in \ker(1 \otimes i)$
  - $u = (\kappa \otimes 1)(u')$ where $\kappa : N \to M$ is the inclusion.
- If R is a domain, and A is a flat R-module, then A is torsion-free.
- Modules over PID are flat iff they are torsion-free.

Tensor products may not commute with direct products:

$$\mathbb{Q} \otimes_{\mathbb{Z}} \prod_{n \geq 2} \mathbb{Z}_n \not\cong \prod_{n \geq 2} (\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}_n)$$

Character Modules

$$B^* = \mathrm{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})$$

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C \iff C^* \xrightarrow{\beta^*} B^* \xrightarrow{\alpha^*} A^*$$

$\implies$ $\mathrm{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ is exact because $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module.

$\impliedby$

$\mathrm{im}\,\alpha \subseteq \ker\beta$. Let $\alpha x \in \mathrm{im}\,\alpha - \ker\beta$, then since $\alpha x \notin \ker\beta$ so $\beta\alpha x \neq 0$.

Exercise 3.19 (i): if G is an abelian group and $a \in G$ is nonzero, there is a homomorphism $f : G \to \mathbb{R}/\mathbb{Z}$ with $f(a) \neq 0$.

Treating C as a $\mathbb{Z}$-module, then there is an $f : C \to \mathbb{Q}/\mathbb{Z} \in \mathrm{Hom}_{\mathbb{Q}}(C, \mathbb{Q}/\mathbb{Z})$ such that $f\beta\alpha(x) \neq 0$. Thus $f \in C^*$ and $f\beta\alpha \neq 0 \implies \beta\alpha = \alpha^*\beta^* \neq 0$. But this contradicts the hypothesis.

$\ker\beta \subseteq \mathrm{im}\,\alpha$.